Math 162
Problem set 6 solutions

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1. (a) We begin with the substitution $u = \sqrt{x}$, which implies that $\mathrm{d}u = \frac{1}{2\sqrt{x}}\,\mathrm{d}x$ and hence

$$\int e^{\sqrt{x}}\,\mathrm{d}x = \int e^u 2\sqrt{x}\,\mathrm{d}u = 2\int ue^u\,\mathrm{d}u\,.$$

Now apply integration by parts: letting $\mathrm{d}v = e^u\,\mathrm{d}u$, we have $v = e^u$ and hence

$$2\int ue^u\,\mathrm{d}u = 2ue^u - 2\int e^u\,\mathrm{d}u = 2ue^u - 2e^u + C = 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C.$$

(b) Let us first compute $\int \sec x\,\mathrm{d}x$, which can be done in several ways. (This was also done in the exam review session, so it's fine if you just cited that.) We start with the observation that

$$\int \sec x\,\mathrm{d}x = \int \frac{\cos x}{\cos^2 x}\,\mathrm{d}x = \int \frac{\cos x}{1-\sin^2 x}\,\mathrm{d}x,$$

then substitute $u = \sin x$ to obtain

$$\int \frac{\cos x}{1-\sin^2 x}\,\mathrm{d}x = \int \frac{\mathrm{d}u}{1-u^2}.$$

The partial fractions decomposition

$$\frac{1}{1-u^2} = \frac{1}{2(1+u)} + \frac{1}{2(1-u)}$$

implies that

$$\int \frac{\mathrm{d}u}{1-u^2} = \int \frac{\mathrm{d}u}{2(1+u)} + \int \frac{\mathrm{d}u}{2(1-u)} = \tfrac{1}{2}\log|1+u| - \tfrac{1}{2}\log|1-u| + C = \tfrac{1}{2}\log\left|\frac{1+u}{1-u}\right| + C.$$

Thus we have

$$\int \sec x\,\mathrm{d}x = \tfrac{1}{2}\log\left|\frac{1+\sin x}{1-\sin x}\right| + C.$$

One can convert this into a more standard form as follows: note that

$$\frac{1+\sin x}{1-\sin x} = \frac{(1+\sin x)^2}{1-\sin^2 x} = \frac{(1+\sin x)^2}{\cos^2 x} = (\sec x + \tan x)^2$$

and hence

$$\tfrac{1}{2}\log\left|\frac{1+\sin x}{1-\sin x}\right| = \log\sqrt{\left|\frac{1+\sin x}{1-\sin x}\right|} = \log|\sec x + \tan x|.$$

One usually finds the answer

$$\int \sec x\,\mathrm{d}x = \log|\sec x + \tan x| + C$$

in integral tables.

As for $\int \sec^3 x \, \mathrm{d}x$, we begin with integration by parts using $u = \sec x$, $\mathrm{d}v = \sec^2 x \, \mathrm{d}x$. Then we have $\mathrm{d}u = \sec x \tan x \, \mathrm{d}x$ and $v = \tan x$, so that

$$\int \sec^3 x \, \mathrm{d}x = \sec x \tan x - \int \sec x \tan^2 x \, \mathrm{d}x \,.$$

Using the Pythagorean identity $1 + \tan^2 x = \sec^2 x$, we obtain

$$\int \sec x \tan^2 x \, \mathrm{d}x = \int \sec^3 x \, \mathrm{d}x - \int \sec x \, \mathrm{d}x = \int \sec^3 x \, \mathrm{d}x - \log|\sec x + \tan x|.$$

Combining this with the above yields

$$\int \sec^3 x \, \mathrm{d}x = \sec x \tan x - \int \sec^3 x \, \mathrm{d}x + \log|\sec x + \tan x|$$

and hence

$$\int \sec^3 x \, \mathrm{d}x = \tfrac{1}{2} \sec x \tan x + \tfrac{1}{2} \log|\sec x + \tan x| + C.$$

(c) Let $u := \arctan x$, i.e. make the substitution $x = \tan u$ where $-\frac{\pi}{2} < u < \frac{\pi}{2}$, so that

$$\sqrt{1+x^2} = \sqrt{1+\tan^2 u} = \sqrt{\sec^2 u} = |\sec u| = \sec u.$$

We have $\mathrm{d}x = \sec^2 u \, \mathrm{d}u$, whence

$$\int \sqrt{1+x^2} \, \mathrm{d}x = \int \sec^3 u \, \mathrm{d}u = \tfrac{1}{2} \sec u \tan u + \tfrac{1}{2} \log|\sec u + \tan u| + C.$$

We have already seen that $\sec u = \sqrt{1+x^2}$, so we obtain that

$$\begin{aligned} \int \sqrt{1+x^2} \, \mathrm{d}x &= \tfrac{1}{2} \sec u \tan u + \tfrac{1}{2} \log|\sec u + \tan u| + C \\ &= \tfrac{1}{2} x\sqrt{1+x^2} + \tfrac{1}{2} \log|x + \sqrt{1+x^2}| + C. \end{aligned}$$

(d) Let $u := \sqrt[6]{x}$, i.e. $x = u^6$ and hence $\mathrm{d}x = 6u^5 \, \mathrm{d}u$. Thus we have

$$\int \frac{\mathrm{d}x}{\sqrt{x} + \sqrt[3]{x}} = \int \frac{6u^5}{u^3 + u^2} \, \mathrm{d}u = \int \frac{6u^3}{u+1} \, \mathrm{d}u \,.$$

Now make the substitution $v = u + 1$, so that $\mathrm{d}v = \mathrm{d}u$ and we obtain

$$\int \frac{6u^3}{u+1} \, \mathrm{d}u = \int \frac{6(v-1)^3}{v} \, \mathrm{d}v = \int (6v^2 - 18v + 18 - \tfrac{6}{v}) \, \mathrm{d}v = 2v^3 - 9v^2 + 18v - 6\log|v| + C.$$

In conclusion, we have

$$\int \frac{\mathrm{d}x}{\sqrt{x} + \sqrt[3]{x}} = 2(1+\sqrt[6]{x})^3 - 9(1+\sqrt[6]{x})^2 + 18(1+\sqrt[6]{x}) - 6\log(1+\sqrt[6]{x}) + C.$$

2. Note the factorization

$$x^4 + 1 = (x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1).$$

(It's easier to find such a factorization if you first observe that $x^4 + 1 > 0$ for all $x \in \mathbb{R}$, which implies that $x^4 + 1$ must factorize into two irreducible quadratic polynomials.) Next, we must find the partial fraction decomposition

$$\frac{1}{x^4+1} = \frac{Ax + B}{x^2 + \sqrt{2}x + 1} + \frac{Cx + D}{x^2 - \sqrt{2}x + 1}.$$

Clearing denominators and grouping like terms, we obtain

$$(A+C)x^3+(-A\sqrt{2}+B+C\sqrt{2}+D)x^2+(A-B\sqrt{2}+C+D\sqrt{2})x+B+D=1.$$

This amounts to the system of four equations in four unknowns

$$\begin{aligned} A+C&=0\\ -A\sqrt{2}+B+C\sqrt{2}+D&=0\\ A-B\sqrt{2}+C+D\sqrt{2}&=0\\ B+D&=1 \end{aligned}$$

Combining the first and third equations yields $B=D$, which together with the fourth equation implies that $B=D=\frac{1}{2}$. Similarly, combining the first, second, and fourth equations yields $A=\frac{1}{2\sqrt{2}}$, $C=-\frac{1}{2\sqrt{2}}$. We conclude that

$$\frac{1}{x^4+1}=\frac{\frac{1}{2\sqrt{2}}x+\frac{1}{2}}{x^2+\sqrt{2}x+1}+\frac{-\frac{1}{2\sqrt{2}}x+\frac{1}{2}}{x^2-\sqrt{2}x+1}=\frac{1}{2\sqrt{2}}\left(\frac{x+\sqrt{2}}{x^2+\sqrt{2}x+1}+\frac{-x+\sqrt{2}}{x^2-\sqrt{2}x+1}\right).$$

Next, observe that the first term can be rewritten

$$\frac{x+\sqrt{2}}{x^2+\sqrt{2}x+1}=\frac{x+\frac{\sqrt{2}}{2}}{x^2+\sqrt{2}x+1}+\frac{\frac{\sqrt{2}}{2}}{x^2+\sqrt{2}x+1}.$$

One of these terms can then be integrated using substitution:

$$\int\frac{x+\frac{\sqrt{2}}{2}}{x^2+\sqrt{2}x+1}\,\mathrm{d}x=\tfrac{1}{2}\int\frac{\mathrm{d}u}{u}=\tfrac{1}{2}\log|u|+C=\tfrac{1}{2}\log(x^2+\sqrt{2}x+1)+C.$$

As for the other, we write

$$\frac{\frac{\sqrt{2}}{2}}{x^2+\sqrt{2}x+1}=\frac{\frac{\sqrt{2}}{2}}{(x+\frac{\sqrt{2}}{2})^2+\frac{1}{2}}=\frac{\frac{\sqrt{2}}{2}}{(x+\frac{\sqrt{2}}{2})^2+(\frac{1}{\sqrt{2}})^2},$$

which can then be integrated as follows:

$$\int\frac{\frac{\sqrt{2}}{2}}{(x+\frac{\sqrt{2}}{2})^2+(\frac{1}{\sqrt{2}})^2}\,\mathrm{d}x=\tfrac{\sqrt{2}}{2}\cdot\sqrt{2}\arctan(\sqrt{2}(x+\tfrac{\sqrt{2}}{2}))+C=\arctan(\sqrt{2}x+1)+C$$

(here we used the formula

$$\int\frac{\mathrm{d}x}{x^2+a^2}=\tfrac{1}{a}\arctan(\tfrac{x}{a})+C,$$

valid for $a>0$). The second term in the partial fraction expansion can be dealt with similarly: first rewrite it as

$$\frac{-x+\sqrt{2}}{x^2+\sqrt{2}x+1}=\frac{-x+\frac{\sqrt{2}}{2}}{x^2-\sqrt{2}x+1}+\frac{\frac{\sqrt{2}}{2}}{x^2-\sqrt{2}x+1},$$

then integrate both terms separately to obtain

$$\int\frac{-x+\frac{\sqrt{2}}{2}}{x^2-\sqrt{2}x+1}\,\mathrm{d}x=-\tfrac{1}{2}\log(x^2-\sqrt{2}x+1)+C$$

and

$$\int\frac{\frac{\sqrt{2}}{2}}{x^2-\sqrt{2}x+1}\,\mathrm{d}x=\arctan(\sqrt{2}x-1)+C.$$

Combining all of the above calculations, we arrive at the final answer

$$\int\frac{\mathrm{d}x}{x^4+1}\,\mathrm{d}x=\frac{1}{2\sqrt{2}}\left(\frac{1}{2}\log\left(\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{2}x+1}\right)+\arctan(\sqrt{2}x+1)+\arctan(\sqrt{2}x-1)\right)+C.$$

3. (a) First we claim that for any $k \geq 0$, there exists a polynomial $p(x)$ such that $f^{(k)}(x) = p(\frac{1}{x})e^{-1/x}$ for all $x \neq 0$. When $k = 0$, we can take $p(x) = 1$. For the inductive step, suppose that for some $k \geq 0$, we have $f^{(k)}(x) = p(\frac{1}{x})e^{-1/x}$. It follows that

$$f^{(k+1)}(x) = -\tfrac{1}{x^2}p'(\tfrac{1}{x})e^{-1/x} + p(\tfrac{1}{x})\tfrac{1}{x^2}e^{-1/x} = \tfrac{1}{x^2}(p(\tfrac{1}{x}) - p'(\tfrac{1}{x}))e^{-1/x},$$

which has the desired form.

In particular f is infinitely differentiable at any $x \neq 0$. As for $x = 0$, we will prove by induction that $f^{(k)}(0) = 0$ for any $k \geq 0$. By definition, we have $f^{(0)}(0) = f(0) = 0$. For the inductive step, suppose that $f^{(k)}(0) = 0$ for some $k \geq 0$. Certainly we have

$$\lim_{h\to 0^-} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} = \lim_{h\to 0^-} \frac{0-0}{h} = 0.$$

On the other hand

$$\lim_{h\to 0^+} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} = \lim_{h\to 0^+} \frac{p(\frac{1}{h})e^{-1/h} - 0}{h} = \lim_{h\to 0^+} \tfrac{1}{h}p(\tfrac{1}{h})e^{-1/h}.$$

Note that $\frac{1}{h}p(\frac{1}{h})e^{-1/h}$ is a sum of terms of the form $\frac{c}{h^n}e^{-1/h}$ where $c \in \mathbb{R}$ and $n \in \mathbb{N}$, and we have

$$\lim_{h\to 0^+} \tfrac{1}{h^n}e^{-1/h} = \lim_{h\to\infty} \frac{h^n}{e^h} = 0.$$

We conclude that

$$\lim_{h\to 0^+} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} = 0$$

and hence $f^{(k+1)}(0) = 0$.

(b) As suggested in the hint, we define $h : \mathbb{R} \to \mathbb{R}$ by

$$h(x) := \frac{f(x)}{f(x) + f(1-x)}.$$

Since f is infinitely differentiable, the chain rule implies that $f(1-x)$ is infinitely differentiable. Moreover, note that $f(x) + f(1-x) \neq 0$ for any $x \in \mathbb{R}$, since $f(x) > 0$ for $x > 0$ and $f(1-x) > 0$ for $x < 1$. Thus h is infinitely differentiable by the quotient rule.

Finally, we define $g : \mathbb{R} \to \mathbb{R}$ by

$$g(x) := h\left(\frac{x-c}{a-c}\right) h\left(\frac{d-x}{d-c}\right).$$

The chain and product rules imply that g is infinitely differentiable because h is, hence (i). For (ii), note that if $a \leq x \leq b$, then $\frac{x-c}{a-c}, \frac{d-x}{d-c} \geq 1$, so that

$$g(x) = h\left(\frac{x-c}{a-c}\right) h\left(\frac{d-x}{d-c}\right) = 1 \cdot 1 = 1.$$

If $x \leq c$ or $x \geq d$ then $\frac{x-c}{a-c} \leq 0$ or $\frac{d-x}{d-c} \leq 0$ respectively, whence

$$g(x) = h\left(\frac{x-c}{a-c}\right) h\left(\frac{d-x}{d-c}\right) = 0$$

and we have verified (iii).

4. (a) First, we prove that the improper integral $\int_1^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ converges. By the Archimedean property we have $x-1 \le n$ for some $n \in \mathbb{Z}$ with $n \ge 0$, and we claim that $\int_1^\infty t^n e^{-t}\,\mathrm{d}t$ converges for all such n. We induct on n, so first notice that when $n=0$ we have

$$\int_1^\infty e^{-t}\,\mathrm{d}t = -e^{-t}\Big|_1^\infty = \frac{1}{e}.$$

For the inductive step, use integration by parts to see that

$$\int_1^\infty t^n e^{-t}\,\mathrm{d}t = -t^n e^{-t}\Big|_1^\infty + n\int_1^\infty t^{n-1}e^{-t}\,\mathrm{d}t = \frac{1}{e} + n\int_1^\infty t^{n-1}e^{-t}\,\mathrm{d}t,$$

where we used the fact that $\lim\limits_{t\to\infty} \frac{t^n}{e^t} = 0$. The latter integral converges by the inductive hypothesis, whence the claim. Since $0 \le t^{x-1}e^{-t} \le t^n e^{-t}$ for all $t \ge 1$, the convergence of $\int_1^\infty t^{x-1}e^{-t}\,\mathrm{d}t$ follows (we have previously made this kind of argument, using the observation that $\int_1^N t^{x-1}e^{-t}\,\mathrm{d}t$ is an increasing and bounded function of N).

It remains to prove the convergence of $\int_0^1 t^{x-1}e^{-t}\,\mathrm{d}t$, which is an improper integral for $x<1$, since in that case the integrand is unbounded on $(0,1]$. Note that

$$\int_0^1 t^{x-1}\,\mathrm{d}t = \lim_{a\to 0^+} \frac{t^x}{x-1}\Big|_{t=a}^{t=1} = \frac{1}{x-1}\lim_{a\to 0^+}(1-a^x) = \frac{1}{x-1}.$$

Here we used the observation that for $x>0$, we have

$$\lim_{a\to 0^+} a^x = \lim_{a\to 0^+} \exp(x\log a) = 0$$

because $\lim\limits_{a\to 0^+} x\log a = -\infty$. We have $e^{-t} \le 1$ for all $t \ge 0$, and hence $0 \le t^{x-1}e^{-t} \le t^{x-1}$, which implies the convergence of $\int_0^1 t^{x-1}e^{-t}\,\mathrm{d}t$.

(b) As suggested, we apply integration by parts to obtain

$$\Gamma(x+1) = \int_0^\infty t^x e^{-t}\,\mathrm{d}t = -t^x e^{-t}\Big|_0^\infty + x\int_0^\infty t^{x-1}e^{-t}\,\mathrm{d}t = -\lim_{t\to\infty}\frac{t^x}{e^t} + x\Gamma(x).$$

Then we observe that

$$\lim_{t\to\infty} x\log t - t = \lim_{t\to\infty} t\left(x\frac{\log t}{t} - 1\right) = -\infty$$

because $\lim\limits_{t\to\infty}\frac{\log t}{t} = 0$, and hence

$$\lim_{t\to\infty}\frac{t^x}{e^t} = \lim_{t\to\infty}\exp(x\log t - t) = 0.$$

(c) Observe that

$$\Gamma(1) = \int_0^\infty e^{-t}\,\mathrm{d}t = -e^{-t}\Big|_0^\infty = 0-(-1) = 1.$$

Thus part (b) implies that for any $n \in \mathbb{N}$, we have

$$\Gamma(n) = (n-1)\Gamma(n-1) = (n-1)(n-2)\Gamma(n-2) = \cdots = (n-1)!\cdot\Gamma(1) = (n-1)!.$$